

CURRICULUM VITAE

NAME-MOHAN TRIVEDI

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Professional Snapshot

Overall 9+ years of experience in IIOT/IOT/Industry 4.0 consulting, Smart factory development, Industrial Automation, Controls, MES development, IT/OT platform for Automotive, EMS Mechanical precision industries. Helping customer to develop and implement digital infrastructure, Industry 4.0 latest technologies and smart solution as paperless factory. Develop enterprise solution architecture for client. Experience and works with global innovative product suppliers to develop smart factory.

Industry 4.0 Management Consultant - Tata Technologies Bengaluru- (August 2020- Till now)

- Assist customer to develop additive manufacturing technology & integrating with MES application. Monitor machine health, machine OK/NG output data for predictive maintenance using data analytics.
- Develop advanced vision with inbuilt processor for manufacturing process quality control. Processing of Vision data and analytics on cloud or on-premise server.
- Industrial automation by Wireless cobot system, equipped with inbuilt vision. Develop a network architecture diagram for industrial cobot pick and place application. Simulate robot as per industrial process requirement to accommodate reachability, turn radius, Robotic arm stress.
- Prepare factory network architecture including IT infra hardware Edge device, PLC, IIOT Sensors, Network Switch by using MQTT, OPC/CA, Profinet, Modbus, TCP IP communication protocol.
- Smart factory layout planning, Process Simulate, BOM, Robotic Simulation and final solution to customer.
- Develop models on virtual cloud (AWS, Azure, google cloud) and on-premise cloud for data processing as a smart industrial solution for any machine.
- Augmented reality solution to making visualize part assembly during assembly process. Creating digital work instruction to help operator execute assembly without mistake. Develop maintenance solution by Augmented reality.
- Smart warehousing system to control and manage material flow and material replenishment by Pick to light system.
- Developing new vendors for developing latest technology of Industry 4.0 and vendor agreement with Request for proposal and request for quotation.
- Extensive experience in delivering End to End digital engineering and manufacturing projects in collaboration with SME, Clients and other stakeholder.
- Develop client specific use cases and mentoring clients to develop Industry 4.0 implementation at their sites.
- Conduct workshop with client & understand main points and potential use case. Develop Industry specific use cases and assist client to develop Industry 4.0 concept at workshop & implement.
- As a part of Digital Manufacturing, Implement Manufacturing Execution system.
- Project track record, Milestones, waterfall study, Capex budget monitoring & approvals.

Senior Process Engineer - Foxconn International Holding, Chennai (April 2018- June 2020)

- Plan, Requirement, budget approval, Procure and validation of product specific and generic Jigs & Fixture for new project.
- Define Process Capability & Process capability Index (Cp & Cpk) of machine for statistical analysis (Gluing machine, thermal Gel Machine, Plasma Machine, CCD machine & Primer Machine).
- Local development & Validation of Jigs & Fixture for project cost reduction as continuous improvement.
- Product yield (productivity) improvement & drive zero defect by advance technology tool and innovation in manufacturing process by POKAYOKE & KAIZEN.
- Work on process and testing automation to reduce human dependency and drive the project for improvements.

- Hands on experience in setup new production line through selecting and validating appropriate jigs, fixture, tool & Equipment.
- Work on Process failure mode effect analysis, Design of Experiment & Control Chart (Statistical process control) with CFT team for improve manufacturing process.
- Help to buyer and sourcing team to develop Local supplier for BOM (Asy. & Packing) & NON BOM material and prepare validation report for customer approval.
- Risk Analysis for the product with short term & long-term action plan.
- Standardize Process & productivity improvement by reducing 8 waste of manufacturing process and measure by UPPH & Yield tools.
- Prepare the Bill of Material for every new project & Change requirement through ECN.
- Develop process flow diagram, Work Instruction, Rework procedure for production line.
- Release the PCF (Product configuration file) matrix to define test tool & SFC tool version.
- Define process routing & configuration to manufacturing Execution system.
- Follow ISO 9000 & 14000 & TQM standard.
- Strong engineering background and provide technical support & guidance to production throughout lifecycle of product.
- Leading a team of 6 members for Industrial Engineering & Process Engineering.

Senior Industrial Engineer - Optiemus Electronics, JV Wistron, Noida (MAY 2017-April 2018).

- Prepare cycle time assessment report (Line balancing) to meet customer forecast.
- Plan for product specific hardware requirement (Jigs, Fixture & Tester) & setup and validation on production line.
- Prepare Ramp up plan for new Product Introduction phase with effective training session to meet customer forecast.
- Production floor layout design with Microsoft Visio & AutoCAD by consider lean manufacturing tool.
- Production Line setup plan (Conveyor & Generic assets requirement from supplier) & execution.
- Time study to reduce waste & benchmark factory UPPH & motion study to recover from fatigue & improve process.
- Design complex test station setup to consider operator ergonomics and improve online maintenance or changeover execution at minimum standard time by SMED tool.
- Analyse operator ergonomics & Fatigue to each assembly & Test stage and plan to eliminate NVA activity.
- Value Stream Mapping by applying lean manufacturing concept to improve throughput time of factory.
- Work on How to eliminating man idle time, machine down time, process scrap for maximum utilization of organization resources.
- Prepare rework costing & ECN costing for management review.
- Productivity improvement by using various engineering tools like UPPH & Yield.
- Create end to end process mapping and process design of the product & interpret through PFD.
- Release working instruction (SOP) to define detail manufacturing process & release to document control system.

Executive Engineer - Videocon Industries Limited, Aurangabad (Jul 2012-Dec 2016).

- Setup & validate the line by DVT & EVT for every new product.
- Line balancing & capacity planning for new project.
- Hands on experience of soldering process of feature phone various component.
- Hands on experience of SAP for release BOM.
- Release SOP & PFD for new product & release for NPI & mass production.
- Top defect analyses by RCA and provide corrective and preventive action.
- Improvement for mass production through poke-yoke & kaizen throughout lifecycle.
- Create & validate the packing line Label design to every new product on Code soft tool.
- Coordinate with R&D team for change into ECN (process, material, Software).

Projects- Tata Electronics Greenfield Site

- Process failure analysis & Corrective Action by statistical analysis & Lean principles to develop and improve process.

- Evaluate Hypothesis test and Design, Run, Validate & concurrence on different DOE
- Process failure analysis through Box plot graph & other QC tools & review with customer.
- Work on large data for Failure analysis & graphical presentation.
- Fishbone & 5 why analysis for RCA & implement corrective action.
- Validate Product ORT requirement & FACA for ORT fail.
- Release DFM, PFC, SOP, Failure Analysis, ERS, FACA for NPI.
- Understanding to study & follow mechanical drawing & MSOP to fulfil process requirements.
- Worked on 5Q machine qualification- Design Qualification, Initial Qualification, Operation Qualification, Production qualification, maintenance Qualification.
- Product testing through Air Leak test, water Leak test, Bubble leakage test & Smart leak test.
- Worked on Product mechanical & vision measurement & testing by UMPT, ISRA, OGP, CMM & Equator.
- Work on Chemical surface treatment process & failure analysis.
- Coordinate with Glue suppliers (Henkel, Atotec, Super X, Cashew) & Validate the Glue Specification, TDS & MSDS.
- Work on CCD Vision system for detecting Glue path, Mechanical Assembly & QC

Project-Smart Manufacturing Lab & Experience centre Setup at Tata Technologies

- Setup Industry 4.0 lab for customer as an experience centre to showcase Industry 4.0. Develop Network architecture to establish communication between Main PLC control system to individual station PLC by MQTT/OPC-UA communication protocol.
- All system data is analysed by on-premise and virtual server by using manufacturing Execution System (MES).
- Setup Smart warehousing system to showcase part replenish message to kitting team. Developing a auto feeding system to assembly station.
- Technologies used- Additive manufacturing, Big data Analytics, AR/VR, Robotic Automation, Process simulation, Robotic Simulation, IIOT, System Integration, Cloud computing, Smart Supply chain control for part replenishment.

Key Skills

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|--------------------------|----------------------|-------------------|----------------------|
| ▪ Digital Twin | ▪ Smart Factory | ▪ Industry4.0 | ▪ IIOT |
| ▪ IT/OT | ▪ Data Analysis | ▪ AR/VR | ▪ Networking |
| ▪ Additive Manufacturing | ▪ Robotic Automation | ▪ Cloud computing | ▪ System Integration |

Certification:

- Six Sigma Black Belt Certification from Council of Six Sigma Certification (CSSC).
- Certification and training in Siemens NX advanced assembly.
- Training & Certification in Lean Concepts, from AIQM.

Academic Credentials:

- Bachelor of Technology in Electronics & Instrumentation branch from Arya Institute of Engineering & Technology, Jaipur.

Personal profiles:

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| ▪ Fathers Name | - Mr. Ravindra Nath Trivedi |
| ▪ Email ID | - mohan.trivedi026@gmail.com |
| ▪ Date of Birth | - 4th April 1991 |
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